

Communication Protocols

ME461 - Group: Little Daisies

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- **Parallel Communication:**

- Multiple wires carry multiple bits simultaneously.
- Fast but bulky and expensive (cables, crosstalk).

- **Serial Communication:**

- Data is sent one bit at a time over a single channel.
- Reduces wiring complexity and cost.
- Standard for most modern embedded systems.



Asynchronous Basics

- **No Clock Line:** Devices must agree on a timing speed (*Baud Rate*) beforehand.
- **Synchronization:** Done via "Start" and "Stop" bits.
- **Common Protocols:** UART (Logic Level), RS-232 (Voltage Level).



Universal Asynchronous Receiver-Transmitter (UART)

- ① **Parallel-to-Serial:** CPU writes byte to Transmit Register → Shift Register pushes bits out one by one.
- ② **Serial-to-Parallel:** Receiver samples line → shifts bits in → CPU reads byte.
- **Wiring:** Cross-coupled (TX connects to RX).
- **Topology:** Strictly Point-to-Point (1 Master, 1 Slave).



UART Frame Structure

Standard 8N1 Frame (8 data bits, No parity, 1 stop bit):

Idle (High)

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Start		D0	D1	D2	D3	D4	D5	D6	D7		Stop
(0)		DATA									(1)

- **Start Bit:** Pulls line LOW (0) to wake up receiver.
- **Data:** Sent LSB (Least Significant Bit) first.
- **Parity (Optional):** Error checking bit.
- **Stop Bit:** Pulls line HIGH (1) to reset for next byte.



ME461 Application: UART Scope Project

Project Context:

- Used for **Raspberry Pi Pico programming** (REPL) and the **Scope Project**.
- The Pico reads signals/PWM and transmits data to the PC for visualization.

Configuration:

- **Baud Rate:** 115200
- **Connection:** UART-over-USB or physical UART pins.

Pico UART Pinout

Function	Pin
TX	GP0
RX	GP1
GND	Any GND

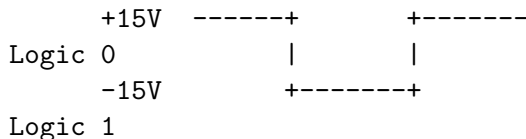
**Scope project sends PWM values to Python script via this link.*



RS-232: The Physical Layer

UART is a logic protocol; RS-232 is the voltage standard often used with it.

- **Purpose:** Long distance communication (up to 15m) where 5V signals would degrade.
- **Voltage Levels (Inverted Logic):**
 - Logic 0: +3V to +15V
 - Logic 1: -3V to -15V



Note: You need a MAX232 converter to connect this to a microcontroller.



SPI: Serial Peripheral Interface

Overview:

- Synchronous: Uses a dedicated clock line (SCK).
- Architecture: Master-Slave.
- Duplex: **Full Duplex** (Send and Receive simultaneously).

Wiring (4 Lines):

- ① **SCK**: Serial Clock (generated by Master).
- ② **MOSI**: Master Out Slave In (Data to slave).
- ③ **MISO**: Master In Slave Out (Data to master).
- ④ **CS/SS**: Chip Select (Active Low, selects the specific slave).



- **Standard Mode:** One CS line per Slave.
- **Daisy Chain:** Slaves connected in series (Output of one goes to Input of next).

Advantages:

- Fastest serial protocol (10MHz - 100MHz).
- No complex addressing (hardware selection via CS).

Disadvantages:

- Requires many wires ($3 + N$ slaves).
- No hardware acknowledgment (blind transmission).



SPI Timing Modes

SPI has 4 modes based on Clock Polarity (CPOL) and Phase (CPHA).

- **CPOL:** Idle state of clock (High or Low).
- **CPHA:** Data sampled on Rising or Falling edge.

Mode 0 (Standard):

SCK: $\overline{} \quad \overline{} \quad \overline{} \quad \overline{}$
 / _ / _ / _ / _ (Sample on Rising Edge)

MOSI: Valid Data



Project Context:

- Used in the **Tetris Game** project.
- High-speed communication required to update the LED grid matrix smoothly.

Hardware:

- **Master:** Raspberry Pi Pico
- **Slave:** MAX7219 Dot Matrix Driver

Pico SPI0 Pinout

Function	Pin
MOSI	GP19
SCK	GP18
CS (SS)	GP17
VCC	5V (VBUS)

**MISO is not used here
(Display only receives data).*



I²C: Inter-Integrated Circuit

Overview:

- Synchronous, Half-Duplex.
- **2 Wires Only:** SDA (Data) and SCL (Clock).
- **Addressed Based:** No Chip Select lines; devices have a 7-bit address.

Key Features:

- Multiple Masters and Multiple Slaves allowed.
- **Open Drain:** Requires Pull-up Resistors (4.7k Ω).
- **Clock Stretching:** Slave can hold SCL low to pause Master.



I²C Frame Structure

[Image of I2C data packet structure]

[Start] [Address (7bit)] [R/W] [ACK] [Data (8bit)] [ACK] [Stop]

- ➊ **Start Condition:** SDA goes LOW while SCL is HIGH.
- ➋ **Address:** Master broadcasts target address.
- ➌ **ACK/NACK:** Receiver pulls SDA low to acknowledge receipt.
- ➍ **Stop Condition:** SDA goes HIGH while SCL is HIGH.



Advantages:

- Low pin count (just 2 pins for up to 127 devices).
- Hardware flow control (ACK/NACK) ensures data integrity.
- Standard for sensors (IMUs, Temp sensors, EEPROMs).

Disadvantages:

- Slower than SPI (Standard: 100kHz, Fast: 400kHz).
- Complex software implementation compared to SPI.
- Pull-up resistors consume power.



Project Context:

- Used on the **Mechaboard** platform.
- **Bus Sharing:** Two different sensors are controlled using the same single pair of I2C pins.

Devices on Bus:

- ① **MPU6050:** IMU (Accel/Gyro)
- ② **SSD1306:** OLED Display

Pico I2C0 Pinout

Function	Pin
SDA	GP4
SCL	GP5
VCC	3.3V

**Different addresses allow both chips to talk on GP4/GP5 simultaneously.*



Comparison Matrix

Feature	UART / RS-232	SPI	I ² C
Wires	2 (TX, RX)	4 (SCK, MOSI, MISO, CS)	2 (SDA, SCL)
Synchronous	No (Async)	Yes	Yes
Speed	Slow (<115 kbps typical)	Fastest (>10 Mbps)	Medium (100-400 kbps)
Topology	Point-to-Point	Single Master, Multi-Slave	Multi-Master, Multi-Slave
Flow Control	None / Software	None	ACK/NACK
Duplex	Full Duplex	Full Duplex	Half Duplex
Best For	PC ↔ MCU, Debugging	Screens, SD Cards	Sensors, Low Pin Count



How to choose?

- Need simple PC logging or long distance? → **UART / RS-232**
- Need high speed data transfer (Display, Flash Memory)? → **SPI**
- Need many sensors with few pins? → **I²C**

Questions?

